

Outline Draft

Loveless, Nix, Stokes, Abernathy's

May 26, 2026

1 Introduction

This serves as an outline for equations of our system of equations for research. Eventually the paper outline may be formed from this document...

2 Equations

Potential systems consist of various mechanisms, but the main differences come from whether or not we contain a Holling-Type III term in our A_β compartment. The other difference comes from whether or not we include a logistic growth term in each compartment [3], [2]. The Equations including the Holling-Type III term is given by

$$\frac{dA_\beta}{dt} = \lambda_{A_\beta} A_\beta (K_{A_\beta} - A_\beta) + \lambda_{A_\beta C a} \frac{Ca^2}{Ca^2 + \sigma^2} - (1 - \alpha_{A_\beta}) \delta_{A_\beta} A_\beta \quad (1)$$

$$\frac{dCa}{dt} = \lambda_{Ca} Ca (K_{Ca} - Ca) + \lambda_{Ca A_\beta} - (1 - \alpha_{Ca}) \delta_{Ca} Ca \quad (2)$$

$$\frac{d\tau}{dt} = \lambda_\tau \tau (K_\tau - \tau) + \lambda_{\tau A_\beta} A_\beta + \lambda_{\tau Ca} - (1 - \alpha_\tau) \delta_\tau \tau \quad (3)$$

$$\frac{dN}{dt} = \lambda_N N (K_N - N) + \lambda_{N\tau} \tau \quad (4)$$

$$\frac{dC}{dt} = \lambda_C C (K_C - C) + \lambda_{CN} NR + \lambda_{C\epsilon} \epsilon. \quad (5)$$

A simpler model can be defined, excluding the Holling-Type III term within dA_β/dt .

$$\frac{dA_\beta}{dt} = a_1 - k_1 A_\beta + \lambda_{A_\beta C a} - (1 - \alpha_{A_\beta}) \delta_{A_\beta} A_\beta \quad (6)$$

$$\frac{dCa}{dt} = a_2 - k_2 Ca + \lambda_{Ca A_\beta} - (1 - \alpha_{Ca}) \delta_{Ca} Ca \quad (7)$$

$$\frac{d\tau}{dt} = a_3 - k_3 \tau + \lambda_{\tau A_\beta} A_\beta + \lambda_{\tau Ca} - \underbrace{(1 - \alpha_\tau) \delta_\tau \tau}_{\text{off the dome}} \quad (8)$$

$$\frac{dN}{dt} = a_4 - k_4 N + \lambda_{N\tau} \tau \quad (9)$$

$$\frac{dC}{dt} = \underbrace{a_5 - k_5 C}_{\text{Pal}} + \underbrace{\lambda_{CN} NR + \lambda_{C\epsilon} \epsilon}_{\text{Petrella}}. \quad (10)$$

In each system, a_n, k_n is the growth and wash out, λ_{ab} is the growth rate for a with respect to b , $ab \neq 0$. α_n is the inefficiency of some drug for n , $0 < \alpha_n < 1$. δ_n is the strength of some drug, $0 < \delta_n < 1$.

The proposed **MODEL TO END ALL MODELS**:

$$\frac{dA_\beta}{dt} = a_1 - r_1 A_\beta + a_2 \frac{Ca^2}{Ca^2 + \sigma^2} \quad (11)$$

$$\frac{dCa}{dt} = b_1 - r_2 Ca + b_2 A_\beta \quad (12)$$

$$\frac{d\tau}{dt} = c_1 - r_3 \tau + c_2 A_\beta + c_3 Ca \quad (13)$$

$$\frac{dN}{dt} = d_1 - k_4 N + d_2 \tau \quad (14)$$

$$\frac{dC}{dt} = e_1 - k_5 C + e_2 NR. \quad (15)$$

where $r_n = k_n + (1 - \alpha_n)\delta_n$.

2.1 Week 3 Equations Currently in work

$$\frac{dA_\beta}{dt} = a_1 - r_1 A_\beta + a_2 \frac{Ca^2}{Ca^2 + \sigma^2} \quad (16)$$

$$\frac{dCa}{dt} = b_1 - r_2 Ca + b_2 A_\beta \quad (17)$$

$$\frac{d\tau}{dt} = c_1 - r_3 \tau + c_2 A_\beta + c_3 Ca \quad (18)$$

$$\frac{dN}{dt} = d_1 \tau \left(1 - \frac{N}{K_N}\right) \quad (19)$$

$$\frac{dC}{dt} = (e_1 N + e_2 \tau) \left(1 - \frac{C}{K_C}\right) \quad (20)$$

and

$$\frac{dA_\beta}{dt} = a_1 - r_1 A_\beta + a_2 \frac{Ca}{Ca + \sigma} \quad (21)$$

$$\frac{dCa}{dt} = b_1 - r_2 Ca + b_2 A_\beta \quad (22)$$

$$\frac{d\tau}{dt} = c_1 - r_3 \tau + c_2 A_\beta + c_3 Ca \quad (23)$$

$$\frac{dN}{dt} = d_1 - k_4 N + d_2 \tau \quad (24)$$

$$\frac{dC}{dt} = e_1 - k_5 C + e_2 NR + e_3 \tau. \quad (25)$$

Var	Value	Unit	Source
A_β	~ 0.01	C	[3]
Ca	0		
τ	0	C	[3]
N	0		
C	0		
a_1	0.25	c/yr	[2]
a_2	2.89	c/yr	[2]
b_1	11	c/yr	[2]
b_2	1000	yr^{-1}	[2]
c_1			
c_2	0.15	yr^{-1}	[1]
c_3			
d_1			
d_2			
e_1	1.67	yr^{-1}	[1]
e_2	3.83	yr^{-1}	[1]
k_1	0.35	yr^{-1}	[2]
k_2	500	yr^{-1}	[2]
k_3			
k_N	1.00	C	
k_C	169.48	C	
α_{A_β}			
α_{Ca}			
α_τ			
δ_{A_β}			
δ_{Ca}			
δ_τ			
σ	0.00027	C	[4]

2.2 Mass Action Terms Model

$$\frac{dA_\beta}{dt} = a_1 - c_1 A_\beta - k_1 A_\beta \tau - \alpha_1 A_\beta \quad (26)$$

$$\frac{dCa}{dt} = b_1 + k_2 A_\beta \tau - c_2 \tau - \alpha_2 \tau \quad (27)$$

$$\frac{d\tau}{dt} = k_3 A_\beta + k_4 \tau - c_3 Ca - \alpha_3 Ca \quad (28)$$

$$\frac{dN}{dt} = rN \left(1 - \frac{N}{K}\right) - k_5 Ca N - k_6 \tau N. \quad (29)$$

3 Parameters

References

- [1] Wenrui Hao, Suzanne Lenhart, and Jeffrey R Petrella. “Optimal anti-amyloid-beta therapy for Alzheimer’s disease via a personalized mathematical model”. In: *PLoS computational biology* 18.9 (2022), e1010481.
- [2] Swadesh Pal and Roderick Melnik. “Nonequilibrium landscape of amyloid-beta and calcium ions in application to Alzheimer’s disease”. In: *Physical Review E* 111.1 (2025), p. 014418.
- [3] Jeffrey R Petrella et al. “Computational causal modeling of the dynamic biomarker cascade in Alzheimer’s disease”. In: *Computational and mathematical methods in medicine* 2019.1 (2019), p. 6216530.
- [4] Yongxin Zhang and Wendi Wang. “Mathematical analysis for stochastic model of Alzheimer’s disease”. In: *Communications in Nonlinear Science and Numerical Simulation* 89 (2020), p. 105347.